developing a single UAV, Skyborg is a modular, open-architecture autonomy framework designed to serve as the “brain” for a wide range of unmanned aircraft. It integrates artificial intelligence (AI), machine learning (ML), and human–machine teaming (HMT) technologies to enable UAVs to think, learn, and operate alongside human pilots across multiple platforms and mission profiles.

The central goal of Skyborg is to create an autonomous, attritable, and low-cost combat drone ecosystem capable of supporting manned fighters in surveillance, strike, and electronic warfare operations. The program introduces the concept of the “Loyal Wingman,” wherein AI-powered UAVs accompany manned aircraft such as the F-35 Lightning II and F-15EX Eagle II, extending sensor coverage, acting as decoys, or engaging in offensive missions independently.

Skyborg’s autonomy core system (ACS) enables drones to perform real-time decision-making, route planning, threat assessment, and target prioritization without direct human control, while still allowing for human oversight. The system continuously learns from mission data, allowing AI algorithms to adapt to evolving combat environments, a capability central to next-generation network-centric warfare (Reilly, 2021).

Several aerospace companies, including Kratos Defense, Boeing, and General Atomics, have contributed UAV platforms like the XQ-58A Valkyrie, UTAP-22 Mako, and MQ-20 Avenger to test and refine the Skyborg AI framework. These collaborative experiments have demonstrated autonomous formation flying, sensor fusion, and swarm coordination, validating the system’s scalability and cross-platform adaptability.

Strategically, the Skyborg Program represents the technological foundation of the US Air Force’s Next Generation Air Dominance (NGAD) initiative, aimed at achieving manned–unmanned teaming (MUM-T) and